

CoreOps Week 2 Wed Module

Kubernetes Internals & High Availability Architecture

Objective: Master control plane internals, node lifecycle, upgrade patterns, HA design, static pod behaviour, and failure-handling logic.

WEDNESDAY: kubelet, kube-proxy, admission webhooks

Critical Concepts

- Kubelet lifecycle
- Pod lifecycle: Pending → Scheduled → ContainerCreating → Running
- CRI behaviour (containerd)
- kube-proxy: iptables & IPVS
- Admission webhooks: Validating vs Mutating
- How the API server talks to webhooks

Key Outcomes

- You can debug Pending vs ContainerCreating vs ImagePullBackOff
- You understand service routing and iptables flows
- You can fix webhook failures that block cluster-wide deployments

WHAT TO PRACTICE (HANDS-ON)

Lab 1: Kubelet failure simulation

- `sudo systemctl stop kubelet`
- `kubectl get nodes`

Lab 2: Flush iptables to break networking

- `sudo iptables -F`
- `kubectl get svc -A`

Observe kube-proxy recreation.

Lab 3: Create & break admission webhook

Deploy a validating webhook → shut it down → try applying manifests.

WHERE TO STUDY

- Kubelet: <https://kubernetes.io/docs/reference/command-line-tools-reference/kubelet/>
- kube-proxy: <https://kubernetes.io/docs/concepts/services-networking/service/>
- Webhooks: Official K8s docs + Open Policy Agent blog

INFRATHRONE MINDSET NOTES

- If kubelet is healthy, your node is healthy; nothing else matters.
 - Service routing failures often come from kube-proxy, not networking.
 - Webhooks can bring your entire cluster down; treat them with respect.
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